

**Class XII**  
**SUBJECT : MODERN COMPUTER APPLICATION(COMA)**

**Total Contact Hours: 200 (Theory & Practical: 180 ; Remedial & Home Assignment:20)**

**SEMESTER – III**

**Course Code: COMA(Theory)**

**Full Marks: 35**

**Contact Hours: 100**

| Unit – 1 | Python Programming                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 25 Marks | Total 80 Hours |
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|          | <ul style="list-style-type: none"> <li><b>Familiarization with the basics of Python programming</b> <ul style="list-style-type: none"> <li>➤ Introduction to Python, Features of Python, executing a simple "hello world" program, execution modes: interactive mode and script mode, Python character set, Python tokens (keyword, identifier, literal, operator, punctuator), variables, concept of l-value and r-value, use of comments.</li> </ul> </li> </ul> |          | <b>4 Hours</b> |
|          | <ul style="list-style-type: none"> <li><b>Knowledge of data types</b> <ul style="list-style-type: none"> <li>➤ Number (integer, floating point, complex), Boolean, sequence (string, list, tuple), None, Mapping(dictionary), mutable and immutable data types.</li> </ul> </li> </ul>                                                                                                                                                                             |          | <b>2 Hours</b> |
|          | <ul style="list-style-type: none"> <li><b>Operators</b> <ul style="list-style-type: none"> <li>➤ Arithmetic operators, relational operators, logical operators, assignment operators, augmented assignment operators, identity operators (is, is not), membership operators (in not in).</li> </ul> </li> </ul>                                                                                                                                                    |          | <b>3 Hours</b> |
|          | <ul style="list-style-type: none"> <li><b>Expressions, statement, type conversion, and input/output</b> <ul style="list-style-type: none"> <li>➤ Precedence of operators, expression, evaluation of an expression, type-conversion (explicit and implicit conversion), accepting data as input from the console and displaying output.</li> </ul> </li> </ul>                                                                                                      |          | <b>3 Hours</b> |
|          | <ul style="list-style-type: none"> <li><b>Errors: Syntax errors, logical errors, and run-time errors.</b></li> </ul>                                                                                                                                                                                                                                                                                                                                               |          | <b>2 Hours</b> |
|          | <ul style="list-style-type: none"> <li><b>Flow of Control</b> <ul style="list-style-type: none"> <li>➤ Introduction, use of indentation, sequential flow, conditional and iterative flow.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                   |          | <b>4 Hours</b> |
|          | <ul style="list-style-type: none"> <li><b>Conditional statements</b> <ul style="list-style-type: none"> <li>➤ if, if-else, if-elseif-else.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                  |          | <b>5 Hours</b> |
|          | <ul style="list-style-type: none"> <li><b>Iterative Statement</b> <ul style="list-style-type: none"> <li>➤ for loop, range (), while loop, break and continue statements, nested loops.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                     |          | <b>7 Hours</b> |

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|          | <ul style="list-style-type: none"> <li>• <b>Strings</b> <ul style="list-style-type: none"> <li>➤ Introduction, string operations (concatenation, repetition, membership and slicing), traversing a string using loops, built-in functions/methods–len(), capitalize(), title(), lower(), upper(),count(), find(), index(), endswith(), startswith(), isalnum(), isalpha(), isdigit(), islower(), isupper(), isspace(),lstrip(), rstrip(), strip(), replace(), join(), partition(), split()).</li> </ul> </li> </ul>                                                                                                                                               | <b>10 Hours</b> |
|          | <ul style="list-style-type: none"> <li>• <b>Lists</b> <ul style="list-style-type: none"> <li>➤ Introduction, indexing, list operations (concatenation, repetition, membership and slicing), traversing a list using loops, built-in functions/methods–len(), list(), append(), extend(), insert(), count(), index(), remove(), pop(), reverse(), sort(), sorted(), min(), max(), sum()); nested lists.</li> </ul> </li> </ul>                                                                                                                                                                                                                                     | <b>14 Hours</b> |
|          | <ul style="list-style-type: none"> <li>• <b>Introduction to Python modules</b> <ul style="list-style-type: none"> <li>➤ Importing module using ‘import &lt;module&gt;’ and using from statement, importing math module (pi(), sqrt(), ceil(), floor(), pow(), fabs(), sin(), cos(), tan()); random module (random(), randint(), randrange()), statistics module (mean(), median(), mode()).</li> </ul> </li> </ul>                                                                                                                                                                                                                                                | <b>12 Hours</b> |
|          | <ul style="list-style-type: none"> <li>• <b>Functions</b> <ul style="list-style-type: none"> <li>➤ Types of function (built-in functions, functions defined in module, user defined functions), creating user defined function, arguments and parameters, default parameters, positional parameters, function returning value(s), flow of execution, scope of a variable (global scope, local scope).</li> </ul> </li> </ul>                                                                                                                                                                                                                                      | <b>14 Hours</b> |
| Unit – 2 | <b>E-Commerce</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>10 Marks</b> |
|          | <ul style="list-style-type: none"> <li>• <b>An introduction to Electronic Commerce</b> <ul style="list-style-type: none"> <li>➤ What is E-Commerce (Introduction and Definition), Main activities E-Commerce, Goals of E-Commerce, Technical Components of E-Commerce, Functions of E-Commerce, Advantages and disadvantages of E-Commerce, Scope of E-Commerce, Electronic Commerce Applications, Electronic Commerce and Electronic Business (C2C, C2G, G2G, B2G, B2P, B2A, P2P, B2A, C2A, B2B, B2C).</li> <li>➤ Internet, Intranet &amp; Extranet, Role of Internet in B2B Application, Web promotion, Banner, Exchange, Shopping Bots.</li> </ul> </li> </ul> | <b>8 Hours</b>  |
|          | <ul style="list-style-type: none"> <li>• <b>Electronic Payment System</b> <ul style="list-style-type: none"> <li>➤ Introduction, Types of Electronic Payment System, Payment Types, Value Exchange System, Credit Card System, Electronic Fund Transfer, Paperless bill, Modern Payment Cash, Electronic Cash.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                             | <b>6 Hours</b>  |
|          | <ul style="list-style-type: none"> <li>• <b>Internet Marketing</b> <ul style="list-style-type: none"> <li>➤ The PROS and CONS of online shopping, The cons of online shopping, Justify an Internet business, Internet marketing techniques, The E-cycle of Internet marketing, Personalization e-commerce.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                 | <b>6 Hours</b>  |

**Class XII**  
**SUBJECT : MODERN COMPUTER APPLICATION(COMA)**  
**SEMESTER – IV**

**Course Code : COMA (Theory)**

**Full Marks : 35**

**Contact Hours : 80**

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| Unit – 1 | <b>Basic of Operating System</b>                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>10 Marks</b> | <b>Total 20 Hours</b> |
|          | <ul style="list-style-type: none"> <li>• <b>Introduction</b> <ul style="list-style-type: none"> <li>➤ Definition</li> <li>➤ Basic Operating System functions, Types of Operating System — Batch processing, Multi-programming, Time sharing, Multi-processing, Distributed and Real-time system.</li> </ul> </li> </ul>                                                                                                                                    |                 | <b>2 Hours</b>        |
|          | <ul style="list-style-type: none"> <li>• <b>Operating System Organization</b> <ul style="list-style-type: none"> <li>➤ Processor and User modes, Kernels, System calls and system programs.</li> </ul> </li> </ul>                                                                                                                                                                                                                                         |                 | <b>2 Hours</b>        |
|          | <ul style="list-style-type: none"> <li>• <b>Process Management</b> <ul style="list-style-type: none"> <li>➤ System view of process and resources, Process Control Block, I/O and CPU bound process.</li> <li>➤ Scheduling : Non-Pre-emptive and Pre-emptive scheduling. Basic concept of First-Come-First Serve (FCFS), Shortest-Job-First (SJF) and Shortest-Remaining-Time-First Scheduling.</li> <li>➤ Deadlock : Basic concepts</li> </ul> </li> </ul> |                 | <b>8 Hours</b>        |
|          | <ul style="list-style-type: none"> <li>• <b>Memory Management</b> <ul style="list-style-type: none"> <li>➤ Physical and Logical address space, Memory allocation strategies, Fixed and Variable partitions, Basic concept of Paging and Segmentation, Virtual memory.</li> </ul> </li> </ul>                                                                                                                                                               |                 | <b>8 Hours</b>        |
| Unit – 2 | <b>Database Management System</b>                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>15 Marks</b> | <b>Total 35 Hours</b> |
|          | <ul style="list-style-type: none"> <li>• <b>Introduction</b> <ul style="list-style-type: none"> <li>➤ Drawbacks of Legacy System, Advantages of DBMS, Layered Architecture of Database, Data Independence, Data Models, Schemas and Instances, Database Languages, Database Users, DBA, Data Dictionary.</li> </ul> </li> </ul>                                                                                                                            |                 | <b>3 Hours</b>        |
|          | <ul style="list-style-type: none"> <li>• <b>Entity Relationship (ER) Modeling</b> <ul style="list-style-type: none"> <li>➤ Entity, Attributes and Relationship, Structural Constraints, Keys (Super Key, Key, Candidate Key, Alternate Key, Primary Key, Foreign Key), ER Diagram of Some Example Database, Weak and strong Entity Set.</li> </ul> </li> </ul>                                                                                             |                 | <b>7 Hours</b>        |
|          | <ul style="list-style-type: none"> <li>• <b>Relational Model</b> <ul style="list-style-type: none"> <li>➤ Basic Concepts of Relational Model, Relational Algebra.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                   |                 | <b>8 Hours</b>        |
|          | <ul style="list-style-type: none"> <li>• <b>Integrity Constraints</b> <ul style="list-style-type: none"> <li>➤ Domain Constraints, Referential Integrity, View.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                     |                 | <b>2 Hours</b>        |

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|          | <ul style="list-style-type: none"> <li>• <b>SQL</b> <ul style="list-style-type: none"> <li>➤ Introduction, Data Definition Language and Data Manipulation Language, Data type (char(n), varchar(n), int, float, date), constraints (not null, unique, primary key), create database, use database, show databases, drop database, show tables, create table, describe table, alter table (add and remove an attribute, add and remove primary key), drop table, insert, delete, select, operators (mathematical, relational and logical), aliasing, distinct clause, where clause, in, between, order by, meaning of null, is null, is not null, like, update command, delete command, aggregate functions (max, min, avg, sum, count), group by, having clause.</li> </ul> </li> </ul> | <b>15 Hours</b> |
| Unit – 3 | <b>Foundation of Artificial Intelligence (AI)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>10 Marks</b> |
|          | <ul style="list-style-type: none"> <li>• <b>Introduction to Artificial Intelligence</b> <ul style="list-style-type: none"> <li>➤ Definition and scope of AI.</li> <li>➤ Historical overview and key milestones.</li> <li>➤ Differentiating AI from human intelligence.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>4 Hours</b>  |
|          | <ul style="list-style-type: none"> <li>• <b>AI Subfields and Technologies</b> <ul style="list-style-type: none"> <li>➤ Machine learning: Supervised, unsupervised, and reinforcement learning.</li> <li>➤ Deep learning and neural networks.</li> <li>➤ Natural language processing (NLP) and computer vision.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>10 Hours</b> |
|          | <ul style="list-style-type: none"> <li>• <b>Applications of AI</b> <ul style="list-style-type: none"> <li>➤ AI in finance: Fraud detection, algorithmic trading, and risk assessment.</li> <li>➤ AI in customer service and chatbots.</li> <li>➤ AI in education: Personalized learning and intelligent tutoring systems.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>8 Hours</b>  |
|          | <ul style="list-style-type: none"> <li>• <b>Ethical and Social Implications of AI</b> <ul style="list-style-type: none"> <li>➤ Bias and fairness in AI systems.</li> <li>➤ Impact of AI on employment and the workforce.</li> <li>➤ AI and social inequality.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>3 Hours</b>  |